

Git: First Workflow

```
$ fork  
$ git clone
```

From project history → local work → shared GitHub copies

Where We Are in the Module

Session	Focus	Big idea
Unix Shell	Files and folders	Work with project files from the command line
Local Git	Version control	Save meaningful checkpoints in project history
GitHub	Remote sharing	Share, collaborate, and showcase work online

Today we connect **Git on your computer** with **GitHub online**.

Learning Goals

By the end of this section, you should be able to explain:

1. What Git is for
2. What **modified**, **staged**, and **committed** mean
3. What **local** and **remote** repositories are
4. The difference between **clone** and **fork**
5. How to fork a GitHub repo and clone your fork

What Is Git For?

Git is a tool for tracking project history.

It helps us answer questions like:

- What changed?
- When did it change?
- Why did it change?
- Who changed it?
- Can I go back to an earlier version?

Analogy: Git as a Project Archive

Imagine you are working on a research project.

You do not want 20 files named:

```
analysis_final.R  
analysis_final_revised.R  
analysis_final_revised_REAL_FINAL.R
```

Instead, Git lets you keep **one project folder** with a clear history of saved checkpoints.

Git in One Sentence

Git lets you save labeled checkpoints as your project changes.

Each checkpoint is called a **commit**.

A commit is like a saved snapshot of your project at a meaningful moment.

The Three States

Git uses three important states for files:

State	Meaning
Modified	You changed the file, but Git has not saved it yet
Staged	You selected the change for the next commit
Committed	Git saved the change in project history

Analogy: Desk → Tray → Archive

Desk

You are actively editing
papers.

Modified

Review Tray

You choose what is ready
to save.

Staged

Archive

You store a labeled
snapshot.

Committed

The Local Git Workflow

Modified on your desk → Staged in review tray → Committed in archive

```
# edit files
```

```
git status      # what changed?  
git add file    # choose this file for the next checkpoint  
git commit -m "message" # save the checkpoint
```

Important: `git add` does **not** save history yet.

It prepares changes for the next commit.

What Does `git status` Do?

`git status` is your checklist.

It tells you:

- which files changed
- which changes are staged
- which changes are not staged
- whether your working tree is clean

```
git status
```

Use it often.

What Are Local and Remote Repositories?

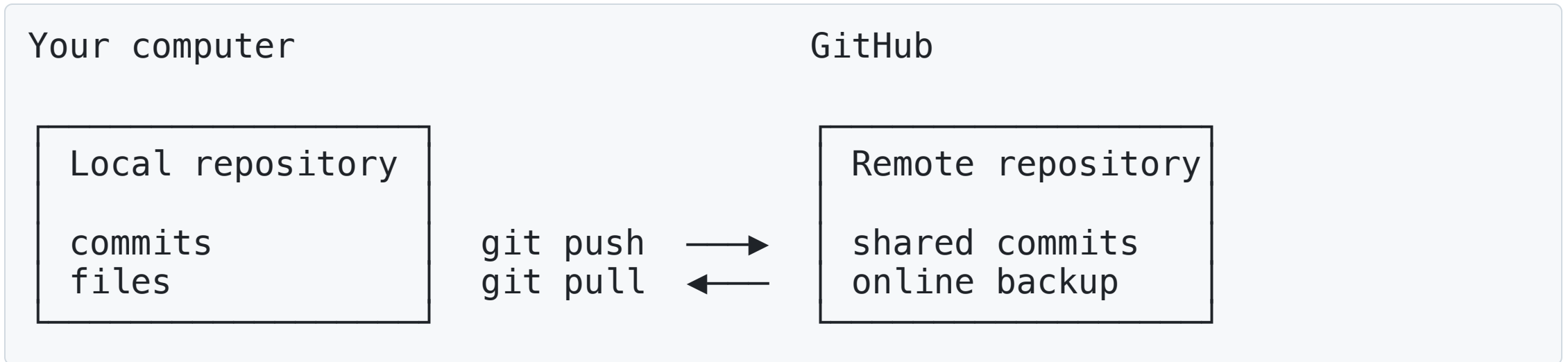
A Git project can have more than one copy.

Copy	Where it lives	Analogy
Local repository	Your computer	Your personal archive
Remote repository	GitHub	Shared online archive

A commit is local first.

GitHub sees it only after you **push**.

Local and Remote: Visual



`push` sends your commits up.

`pull` brings remote commits down.

What Is GitHub?

GitHub is a website that hosts Git repositories online.

It is commonly used for:

- backing up projects online
- collaborating with other people
- reviewing code and changes
- submitting assignments
- building a public portfolio

Git and GitHub are related, but they are not the same thing.

Clone vs Fork

Both are ways to copy a repository, but they copy it to different places.

Action	What it creates	Where
Clone	A local copy	Your computer
Fork	Your own remote copy	Your GitHub account

In this activity, we will usually **fork first**, then **clone our fork**.

Clone: Visual

GitHub original repo
UofT-DSI/git



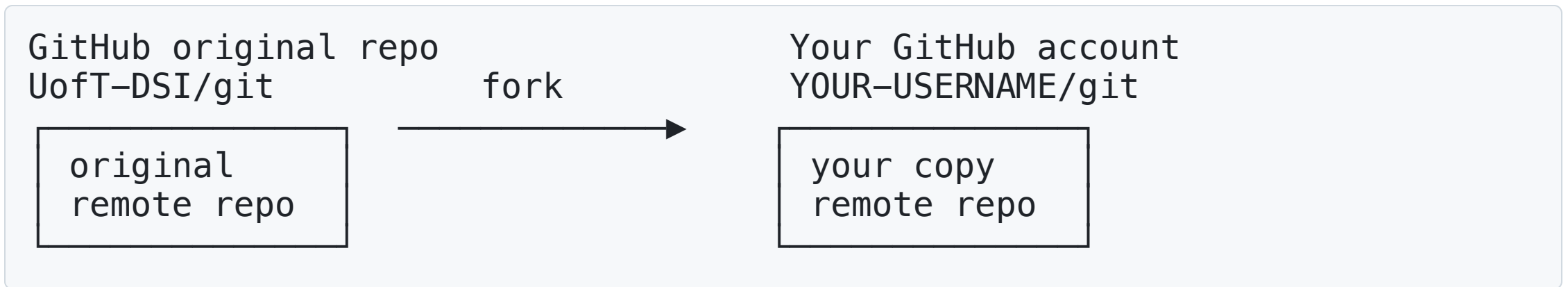
git clone

Your computer
local copy of UofT-DSI/git

Cloning downloads a repository to your computer.

But it does **not** create your own copy on GitHub.

Fork: Visual



Forking creates your own GitHub copy of someone else's repository.

Fork + Clone: Visual

1. Fork on GitHub

UofT-DSI/git  YOUR-USERNAME/git

2. Clone your fork

YOUR-USERNAME/git  your computer

After this, you can work locally and push changes to **your fork**.

Why Fork for This Exercise?

We do not want everyone pushing changes to the original course repository.

Forking gives each learner:

- their own remote copy
- permission to push to their own copy
- a safe place to practice
- a visible GitHub project they can revisit later

Code Along

Fork and clone the course repository.

Step 1: Open the Course Repository

Open this page in your browser:

```
https://github.com/UofT-DSI/git
```

Make sure you are signed in to GitHub.

Then click **Fork**.

Step 2: Create Your Fork

On the fork page:

1. Keep the owner as your GitHub account
2. Keep the repository name as `git`
3. Add a description if you want
4. Click **Create fork**

You should now be on a page like:

```
https://github.com/YOUR-USERNAME/git
```

Step 3: Copy the Clone URL

On your forked repository:

1. Click the green **Code** button
2. Select **HTTPS**
3. Copy the URL

It should look like:

```
https://github.com/YOUR-USERNAME/git.git
```

Step 4: Clone Your Fork

In your terminal, move to the folder where you keep class work.

Then run:

```
git clone https://github.com/YOUR-USERNAME/git.git
```

Replace `YOUR-USERNAME` with your GitHub username.

Step 5: Enter the Repository

```
cd git
```

Check where you are:

```
pwd  
ls
```

Then ask Git what it sees:

```
git status
```


Step 6: Check the Remote Connection

```
git remote -v
```

You should see something like:

```
origin  https://github.com/YOUR-USERNAME/git.git (fetch)  
origin  https://github.com/YOUR-USERNAME/git.git (push)
```

`origin` is the default name for the remote repository you cloned from.

Stop and Check

Can you answer these questions?

1. Where is the original repository?
2. Where is your fork?
3. Where is your local clone?
4. What does `origin` point to?
5. Why did we fork before cloning?

Common Mistakes

Problem	What to check
Cloned the original repo	Did your URL include your username?
Cannot push later	Did you clone your fork or the original?
Cannot find the folder	Did you run <code>cd git</code> ?
Not signed in	Are you signed in to GitHub before forking?
Wrong folder	Run <code>pwd</code> and <code>ls</code> to orient yourself

Key Takeaways

- Git tracks project history with commits.
- Files move from **modified** → **staged** → **committed**.
- Local repositories live on your computer.
- Remote repositories live online, often on GitHub.
- Clone copies a repo to your computer.
- Fork creates your own GitHub copy.
- In this class: **fork first, then clone your fork**.

References

- Pro Git book: <https://git-scm.com/book/en/v2>
- UofT-DSI Git repository: <https://github.com/UofT-DSI/git>